“AICSYNC” CENTRALIZED MULTI-BRANCH ENROLLMENT & ACADEMIC RECORDS SYSTEM OF ASIAN INSTITUTE OF COMPUTER STUDIES

Undergraduate Capstone Project
Submitted to the Faculty
Cavite State University-Bacoor City Campus
Bacoor City Cavite

In partial fulfillment
of the requirements for the degree of
Bachelor of Science in Information Technology

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An undergraduate capstone project submitted to the faculty of the Department of Computer Studies, Cavite State University, Cavite in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology with Contribution No.______ Prepared under the supervision of Ms. Jovelyn D. Ocampo.

INTRODUCTION

Efficient academic information management is essential for educational institutions to ensure accurate record keeping, organized enrollment procedures, and effective student services. The implementation of a centralized academic information system improves institutional operations by integrating admission, enrollment, academic records management, and reporting into a single platform. Through centralized data management, institutions can enhance accessibility, reduce redundancy, minimize manual errors, and strengthen monitoring across multiple branches.

The Asian Institute of Computer Studies (AICS), established in 1996, continues to enhance its academic services through technological advancement. The institution operates across multiple branches, including the Bacoor Branch located at FYNN Commercial Building, Gen. E. Aguinaldo Highway, Panapaan IV, Bacoor City, Cavite; the General Mariano Alvarez (GMA) Branch located at Tapia Building, Governor's Drive, Brgy. Gavino Maderan, General Mariano Alvarez, Cavite; and the Taytay Branch located at Brgy. San Juan, Taytay, Rizal. With multiple campuses operating under one institution, the use of a centralized system supports better coordination and consistent management of academic data across branches.

Students are also provided with a platform where they can access academic information, submit requirements, and receive institutional updates efficiently. In line with this, a centralized web-based system is introduced to enhance accessibility and support the efficient management of academic services across all branches.

Project Context

Initially, the Asian Institute of Computer Studies (AICS) experienced challenges in managing student enrollment and academic records due to its reliance on manual processes. Activities such as enrollment processing, updating student information, releasing grades, and handling document requests were performed using physical forms and manual encoding. These practices often resulted in slow document retrieval, repeated searching of records, and an increased risk of missing or inconsistent academic information, especially when managing records across multiple branches.

In addition to these challenges, administrative personnel such as branch administrators and registrars encountered difficulties in monitoring student requirements and academic progress. Enrollment requirements were submitted manually through face-to-face transactions, which made verification time-consuming and frequently caused overcrowding within the school. Furthermore, student records were stored separately in each branch, making it difficult and time-consuming to verify enrollment status and consolidate academic data. The absence of an integrated school management system increased the workload of administrative staff and often caused delays in enrollment processing, grade release, and report generation.

Moreover, area managers experienced limitations in monitoring the overall performance of different branches due to the lack of real-time and consolidated data. Tracking the number of enrolled students per course or strand and generating institution-wide reports required manual coordination with each branch. This process delayed decision-making and increased the possibility of data inconsistencies and reporting errors. These challenges highlight the need for a centralized digital system that can efficiently manage enrollment and academic records across all AICS branches.

The system is designed to address these challenges by centralizing enrollment and academic records into a single digital platform. The system allows secure storage and retrieval of student credentials, automated monitoring of enrollment status and academic progress, and role-based access for students, administrators, registrars, and area managers. By eliminating manual record handling and consolidating data across branches, the system improves accuracy, reduces administrative workload, and enhances the overall efficiency and reliability of academic information management within AICS.

Objective of the Study

The general objective of the study is to design and develop a web-based credential-centric academic information system for the Asian Institute of Computer Studies (AICS). The proposed system aims to centralize student enrollment and academic records across multiple branches of the institution in order to improve the efficiency, accuracy, and accessibility of academic information. This objective is justified by the existing challenges encountered by the institution, including manual processing of enrollment, fragmented academic records across branches, and delays in retrieving student credentials and academic reports. Through the development of a centralized digital platform, the study seeks to enhance accessibility to academic information, reduce administrative workload, minimize data inconsistencies, and improve monitoring of student records and institutional operations across AICS branches.

Specifically, the study aimed to:

1.Design the proposed system using an suitable development methodology that will:

a. provide a user-friendly and intuitive web interface for students, administrators, registrars, and area managers;
b. enable efficient management of enrollment, academic records, and student credentials through role-based access;
c. support centralized monitoring of student academic progress and enrollment status across different branches; and
d. address the inefficiencies present in the existing manual enrollment and academic record management processes of AICS.

2. Develop the system using the following technologies:

a. React.js for the front-end to create a responsive and interactive user interface;
b. Django for the back-end to manage system logic, data processing, and database operations; and
c. a centralized database architecture to ensure secure storage, consistency, and reliability of academic records.

3. Test the developed system in terms of functionality:

a. conducting comprehensive system testing;
b. ensuring compatibility across different devices and web browsers;
c. assessing system performance across various screen sizes.

4. evaluate the system using the ISO 25010 software quality standards.

5. prepare an implementation plan.

Purpose and Description

The proposed centralized web-based enrollment and academic records system is developed to improve the management of academic information within the Asian Institute of Computer Studies (AICS). The study aims to provide a digital solution that enhances efficiency, accuracy, and accessibility of enrollment and academic records across multiple branches. It contributes to the improvement of institutional processes by integrating various academic services into a single platform and providing a more organized and systematic approach to data management.

The system offers several capabilities that support academic operations. It provides online enrollment processing, centralized storage of academic records, real-time access to student information, and automated report generation. It also includes role-based access control to ensure data security and proper user authorization. In addition, the system features a centralized dashboard for monitoring academic and enrollment data, making it a relevant and practical solution for multi-branch institutions. The following stakeholders will benefit significantly from the system:

Students can access their enrollment status, academic records, and class schedules through the system. They can also submit requirements online, allowing easier access to academic services.

Administrators can oversee system operations, manage user accounts, and monitor institutional data through a centralized platform, supporting better coordination and control.

Registrars are able to manage student records, validate enrollment requirements, process grades, and generate academic reports efficiently and accurately.

Area Manager can monitor branch performance, enrollment trends, and student academic progress using a centralized dashboard, which supports data-driven decision-making.

Time and Place of the Study

The study is conducted at the Asian Institute of Computer Studies (AICS), with particular focus on the Bacoor Branch and selected branches involved in student enrollment and academic record management. The project started in January 2026, during which the researchers began planning the scope and limitations of the proposed system. The Agile methodology was adopted during this phase to guide the planning process through iterative development and continuous improvement.

In January 2026, the researchers gathered the necessary information regarding the current admission, enrollment, and academic record management processes of the institution. This phase focused on identifying the existing procedures and determining the system requirements that would serve as the foundation for the development of the proposed system.

In February 2026, the researchers will complete the documentation of the capstone project and will begin the initial development of the system. During this period, core features and preliminary system designs will be developed to support user roles such as students, administrators, registrars, and area managers.

In March 2026, revisions will be implemented for both the system and the documentation based on initial evaluations and feedback. At this stage, the researchers will continue refining the system features while ensuring that the development remains aligned with the identified requirements and objectives of the project.

By April 2026, the researchers will continue improving the system by refining its existing features and enhancing the overall workflow of the platform. Adjustments will be made to ensure that the system structure and functionality support the intended admission and enrollment processes.

In May 2026, further development and integration of additional system components will be conducted. Minor revisions will also be applied to both the system and the documentation to maintain consistency and improve the overall functionality of the proposed system.

In June 2026, the system will undergo further testing and validation to ensure functionality, reliability, and readiness for formal evaluation. Preparations for system presentation and defense will also be undertaken during this period.

From July to August 2026, revisions and enhancements will be implemented based on feedback from evaluations and consultations. During this phase, improvements will focus on refining the system features and enhancing the overall usability of the platform.

From September to October 2026, the system development will approach completion. During this stage, final adjustments and refinements will be applied to ensure that all necessary system functions are properly implemented.

In November 2026, comprehensive system testing and evaluation will be conducted to assess the system's usability, effectiveness, and reliability in addressing the identified problems.

In December 2026, final revisions and improvements will be implemented based on evaluation results. This stage will focus on finalizing both the system and the documentation to ensure full compliance with the project requirements.

Scope and Limitation of the Study

The primary focus of the study is the development and implementation of "AICSYNC," a centralized web-based enrollment and academic records system for the Asian Institute of Computer Studies (AICS). The system is designed to replace manual, paper-based workflows with a digital platform that integrates admission, enrollment, and record management into a single repository. By centralizing these processes, the system aims to improve data accuracy and operational efficiency across the institution's multiple branches.

The scope of the system centers on providing a unified platform for four specific user roles, namely the Students, Registrars, Administrators, and Area Managers. For students, the system facilitates the viewing of quarterly grades for Senior High School and semestral grades for College, access to class schedules, and the online submission of enrollment requirements. Additionally, the system includes a module for students to participate in instructor evaluations, ensuring a streamlined way to gather academic feedback.

The administrative and registrar modules are designed to handle the validation of student information, the approval of enrollment applications, and the digital encoding of academic grades. For management oversight, the system provides a centralized dashboard specifically for Area Managers to monitor branch-wide statistics and review consolidated reports in real-time. To ensure the reliability and continuity of these institutional records, the system also incorporates automated backup functions for data security.

Despite its features, the study has several geographical and functional limitations. The AICSYNC system is implemented only in selected branches of the Asian Institute of Computer Studies (AICS), specifically Bacoor, General Mariano Alvarez (GMA), and Taytay within the CALABARZON region. These branches were chosen to ensure manageable system implementation and consistent data handling.

The system does not cover AICS branches outside the CALABARZON region, as these branches operate under different area managers and may follow different systems and procedures. As a result, the proposed system is not designed to integrate with or replace the systems used in those locations.

The system also relies on a stable internet connection to support real-time data updates and smooth data sharing across the selected branches.

In addition, the system is limited to managing academic and administrative data and does not include online payment functionalities. All payment-related transactions will continue to be handled through existing manual processes.

Furthermore, the reporting features available to Area Managers are limited to system-generated summaries and dashboard views and do not support advanced or customizable reports. The evaluation feature is also limited to student-to-instructor feedback and does not cover other types of institutional evaluation.

Conceptual Framework

The conceptual framework of the study used the Input-Process-Output model to illustrate the stages of the project development.

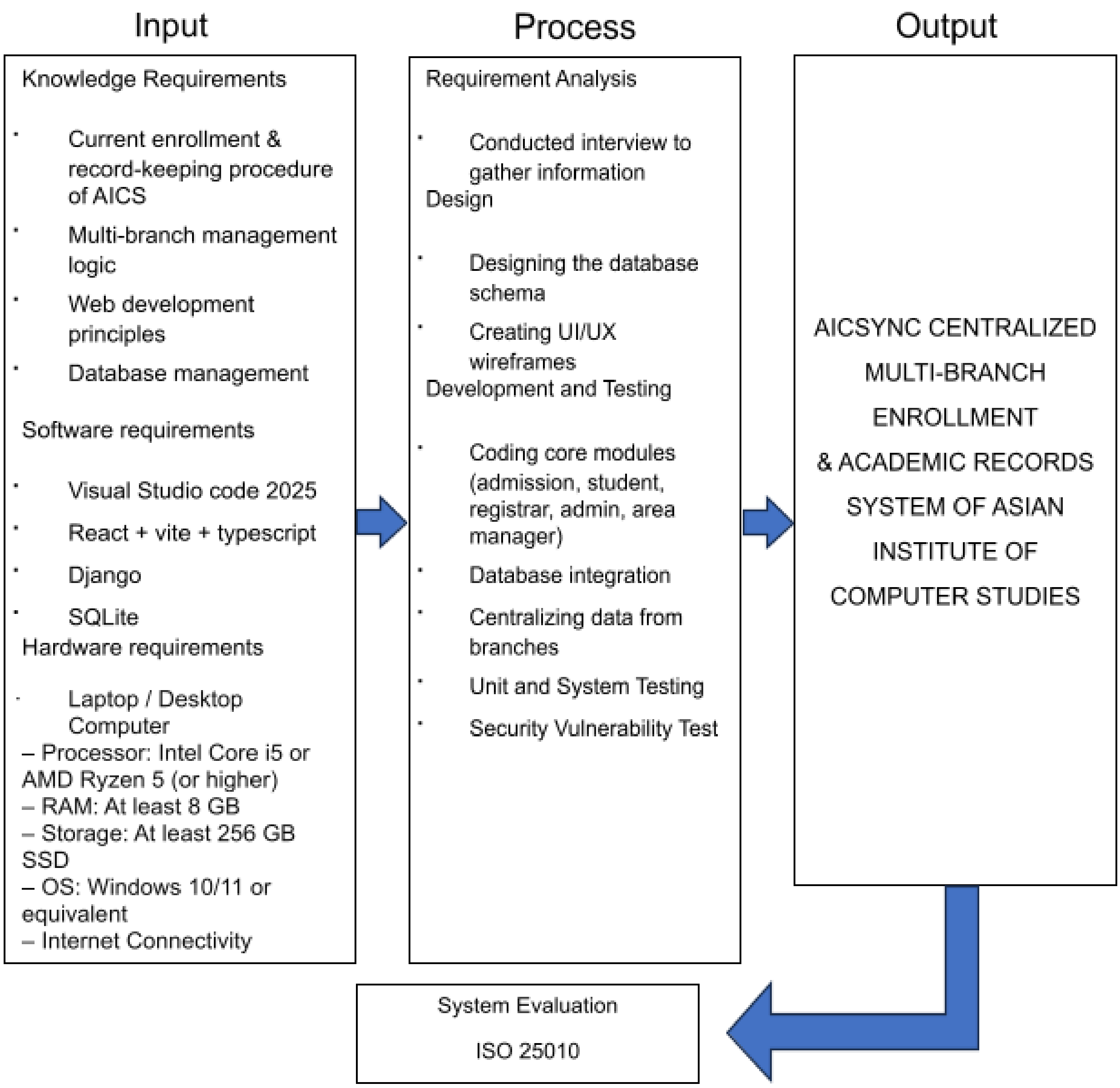

Figure 1 Conceptual Framework

Figure 1 shows the conceptual framework of the study follows the In

The Output Phase represents the developed AICSYNC centralized multi-branch enrollment and academic records system. The system provides a unified platform for managing enrollment, academic records, and institutional data across selected AICS branches, improving efficiency, accuracy, and accessibility of information.

Finally, the system undergoes Evaluation using the ISO 25010 software quality standards to assess its functionality, usability, reliability, performance efficiency, and security. This ensures that the developed system meets quality standards and is suitable for implementation.

Definition of Terms

Academic Information System – A computerized system used by educational institutions to manage student data, enrollment records, grades, and other academic-related information efficiently.

Area Manager – The user responsible for monitoring branch performance, analyzing enrollment trends, and reviewing consolidated academic reports across multiple campuses.

Context Diagram – A high-level diagram that presents the system as a whole and shows its interaction with external entities such as users and other systems.

Data Consolidation – The process of combining and organizing data from multiple branches into a single, unified system to improve accessibility, consistency, and reporting.

DFD (Data Flow Diagram) – A diagram that illustrates the flow of data within a system, showing how data is processed, stored, and transferred between components.

Django – A high-level back-end web framework used to manage system logic, handle data processing, and facilitate secure interaction with the database in web applications.

React – A JavaScript library used for building interactive and responsive user interfaces, particularly for the front-end of web applications.

Role-Based Access – A system feature that restricts or grants access to system functions and data based on the assigned roles of users.

SQLite – A lightweight relational database management system used for storing and managing structured data within the system.

Student Panel – The system interface designed for students where they can view academic records, grades, enrollment status, class schedules, and submit requirements online.

System Integration – The process of connecting different system components into a unified platform to ensure smooth communication and efficient operation.

TypeScript – A superset of JavaScript that adds static typing, improving code reliability, readability, and maintainability in large-scale applications.

UML (Unified Modeling Language) – A standardized modeling language used to visualize, design, and document the structure and behavior of a system.

Use Case Diagram – A type of UML diagram that illustrates how users interact with the system by showing system functionalities and user roles.

Vite – A modern front-end development tool that provides fast build processes and efficient project setup for web applications.

REVIEW RELATED LITERATURE

This chapter presents the related literature and studies that provide the foundation for the development of the proposed system. The reviewed materials include both foreign and local sources that discuss the importance of digital academic information systems, student record management, and web-based enrollment platforms. These studies and literature help establish the relevance of the research and support the need for a centralized academic information system.

Foreign Related Literature

A Student Information System (SIS) is widely used in educational institutions to efficiently manage student data and academic records. According to Das et al. (2023), a Student Information System enables institutions to automate various academic processes such as enrollment, attendance monitoring, and grade management. By digitizing these processes, educational institutions can improve data accuracy and significantly reduce the workload associated with manual record handling.

Student Record Systems (SRS) also play an essential role in organizing and managing academic information. Simwamba (2025) explained that digital record management systems minimize data redundancy and enhance accessibility while ensuring the secure storage of student information. Compared to traditional paper-based methods, digital systems provide faster retrieval of records and improved data integrity.

Web-based academic systems further improve accessibility and communication within educational institutions. Smith and Brown (2023) stated that online academic platforms allow students, faculty members, and administrators to access academic records in real time. This capability promotes better coordination, transparency, and efficiency in managing institutional information.

Records management systems are essential for handling large volumes of institutional data. Fernandez et al. (2025) emphasized that digital records management systems enhance data storage, retrieval efficiency, and security. These systems are more reliable than traditional paper-based methods because they reduce the risk of data loss and improve the overall management of academic information.

Educational technology systems also contribute significantly to institutional efficiency. CooperGibson Research (2022) highlighted that digital technologies improve administrative processes, facilitate better data organization, and support informed decision-making in educational institutions when implemented effectively.

Local Related Literature

Academic information systems are increasingly being implemented in Philippine educational institutions to improve administrative processes and data management. The University of the Philippines developed the **Student** Academic Information System (SAIS), which manages enrollment, grading, and academic records in a centralized digital platform (UP System, 2022). This system demonstrates how centralized academic systems can enhance operational efficiency and improve accessibility to academic information.

Records management systems also help institutions handle increasing volumes of academic data more effectively. Angala et al. (2023) explained that digital document management systems significantly improve the storage, retrieval, and organization of institutional records compared to traditional paper-based methods.

Enrollment systems play an important role in streamlining student registration processes. Ali (2022) stated that web-based enrollment systems reduce delays in the registration process while improving the accuracy of student data and overall management of academic records.

Digital registrar systems improve the delivery of student services. Adra et al. (2024) emphasized that online records management systems enable educational institutions to process student documents more efficiently and provide faster services to students.

Local institutions continue to adopt digital systems to enhance operational efficiency. Rose (2025) found that student records systems significantly improve document storage, retrieval efficiency, and overall data management while maintaining compliance with software quality standards.

Foreign Related Studies

Karpe et al. (2022) developed a web-based academic performance tracking system that integrates student records, attendance monitoring, and assessments into a centralized platform. The study showed that such systems improve monitoring of student performance and support better decision-making in educational institutions.

Ramani et al. (2025) designed a Student Record System that automates the storage and retrieval of student information. Their findings revealed that the system significantly reduced manual errors and improved the overall efficiency of administrative operations.

Afolabi et al. (2023) developed an electronic learner data management system that enhances data accuracy, accessibility, and security through centralized digital storage. The system allows educational institutions to manage student data more effectively while maintaining data integrity.

Arciso and Casquejo (2025) implemented an electronic learner management system using the Agile development methodology. Their findings showed that the system achieved high user acceptance due to its ease of use and improved efficiency in handling academic information.

Silwamba and Matela (2025) developed a digital student records system designed to improve institutional data management. Their study found that the system reduced data redundancy and enhanced the operational efficiency of educational institutions.

Local Related Studies

Purcia and Velarde (2022) conducted a study on student registration systems and found that the digitization of academic processes significantly improves accessibility and efficiency in managing student records.

Nisperos and Eustaquio (2023) developed a web-based student registration system with decision-support features. The system improved usability and operational efficiency by providing better data management and automated processing of student information.

The University of the Philippines (2024) evaluated the Student Academic Information System (SAIS) and found that the system improved data accuracy, reduced processing delays, and enhanced academic monitoring across different campuses.

Grepon et al. (2021) developed a School Management Information System using the Agile methodology. Their findings showed that the system improved data handling processes and increased operational efficiency within educational institutions.

Campanan et al. (2024) created a student information system for senior high schools that improved data accuracy, transparency, and communication between schools and parents.

System Technical Background

The proposed AICSYNC system is a web-based centralized enrollment and academic records management system designed for multi-branch educational institutions. It follows a client-server architecture, where users access the system through web browsers. The front-end is developed using React.js to provide a responsive user interface, while the back-end uses Django for processing, authentication, and database interaction. The system utilizes a centralized database to ensure data consistency across branches. Role-based access control is implemented to manage user permissions for students, registrars, administrators, and managers. Algorithms for data validation, scheduling, and record retrieval ensure efficient system performance and accuracy.

System Architecture

The AICSYNC system follows a client-server architecture in which users access the system through a web browser over the internet. The system is composed of three main components: the front-end, back-end, and database, which work together to support centralized academic information management across selected AICS branches.

The front-end is developed using React.js and is responsible for handling user interface and interactions, allowing users to input data, navigate the system, and view academic information. The back-end is developed using Django, which processes user requests, manages system logic, handles authentication, and communicates with the database. The database serves as the central storage for all academic records, enrollment data, and user information, ensuring data consistency and real-time access across branches. Through this architecture, the system enables efficient, secure, and accurate communication between users and system components.

Synthesis

The reviewed literature and studies highlight the increasing adoption of digital systems for managing student information, enrollment processes, and academic records in educational institutions. Both foreign and local literature consistently indicate that traditional manual systems are inefficient, prone to human error, and difficult to manage, particularly when handling large volumes of academic data. As educational institutions expand and the demand for efficient information management increases, the need for reliable digital academic systems becomes more evident.

Foreign literature such as those by Das et al. (2023), Simwamba (2025), and Smith and Brown (2023) emphasize the importance of Student Information Systems (SIS) and digital record management systems in improving data accuracy, accessibility, and administrative efficiency. These studies highlight the role of web-based platforms in automating academic processes, including enrollment, attendance tracking, and grade management. Similarly, Fernandez et al. (2025) and CooperGibson Research (2022) explain that digital record management and educational technologies contribute to improved institutional operations by strengthening data organization, enhancing security, and supporting informed decision-making. These foreign studies primarily focus on the technological benefits of centralized systems and the role of digital transformation in academic management.

Local literature also supports the implementation of digital academic systems in educational institutions. The Student Academic Information System (SAIS) developed by the University of the Philippines (UP System, 2022) demonstrates how centralized academic platforms can effectively manage enrollment, grading, and academic records within a university setting. In addition, Angala et al. (2023) and Ali (2022) highlight that web-based systems improve document organization and streamline student registration processes. Similarly, Adra et al. (2024) and Rose (2025) emphasize that digital registrar systems and student record management platforms enhance service delivery, improve accessibility to academic data, and increase administrative efficiency. Compared with foreign literature, local studies tend to focus more on the practical implementation of digital systems within specific educational institutions in the Philippines.

The reviewed foreign and local studies further demonstrate the effectiveness of web-based academic management systems. Karpe et al. (2022), Ramani et al. (2025), and Afolabi et al. (2023) developed systems that integrate student records, academic performance monitoring, and data storage into centralized digital platforms. Their findings indicate that such systems reduce manual errors and significantly improve the management and accessibility of academic information. Additionally, Arciso and Casquejo (2025) and Silwamba and Matela (2025) emphasize the role of digital learner management systems and Agile-based development approaches in improving system usability and institutional efficiency.

Similarly, local studies conducted by Purcia and Velarde (2022), Nisperos and Eustaquio (2023), and Campanan et al. (2024) demonstrate that web-based registration and student information systems improve transparency, communication, and efficiency in managing academic data. Grepon et al. (2021) also revealed that implementing school management systems using Agile methodologies enhances data handling processes and operational efficiency within educational institutions. These studies collectively highlight that the use of modern software development methodologies and web technologies contributes significantly to the effectiveness of academic information management systems.

Despite the similarities in the objectives of these systems, most of the reviewed studies focus primarily on managing academic records within a single institution or campus. While these systems effectively address issues related to manual data management, they often lack the capability to integrate academic records across multiple branches of an institution. This limitation highlights the need for a more comprehensive system capable of supporting centralized data management across multiple campuses. Such a system would enable better coordination among branches, improve accessibility of academic records, and enhance institutional efficiency in managing enrollment and academic information.

METHODOLOGY

This chapter includes the design of the software, systems, products, or process, system development, system testing and evaluation, data analysis and the implementation plan of the study.

Design of Software, System, and Processes

The development of AICSYNC: Centralized Multi-Branch Enrollment & Academic Records System of the Asian Institute of Computer Studies aims to improve the management of enrollment and academic records through a centralized web-based platform. The system addresses the institution's reliance on manual document filing, slow retrieval of student credentials, manual backtracking of alumni records, and overcrowded physical files.

The proposed system provides major features such as online admission registration, student credential storage, grade viewing, enrollment status monitoring, document request processing (TOR, COR, COG, Form 137), administrative dashboard, and backup and archive management.

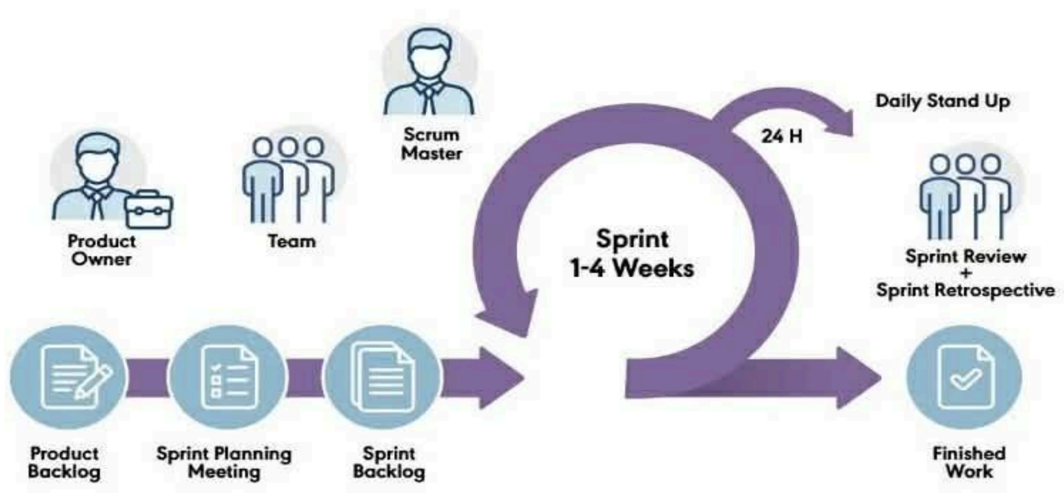

Figure 2. Scrum Methodology (Sutherland & Schwaber 1995)

Scrum methodology is a framework under the Agile approach that focuses on iterative and incremental software development. It emphasizes collaboration, flexibility, and continuous delivery of functional software. In Scrum, work is divided into short cycles called sprints (usually 1–4 weeks), with clearly defined roles—Product Owner, Scrum Master, and Developers—and structured events such as Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective. Scrum helps teams respond quickly to changing requirements, deliver working increments frequently, and continuously improve both the product and the development process.

Guided by the Scrum methodology, the researchers first gathered all necessary information regarding the current enrollment and academic record processes at the Asian Institute of Computer Studies (AICS). Key challenges identified include manual enrollment processing, inefficient record monitoring, and poor credential and data management.

Based on the gathered information, the researchers identified the key features and functional requirements necessary for the proposed system, including a Student Panel for viewing grades and class schedules, an Admin Panel for managing student and alumni records, a Registrar module for validating student information and approving enrollment, and an Area Manager dashboard for monitoring branch performance. These requirements were compiled into the Product Backlog, which contained all proposed features, system improvements, and functional requirements for the web-based AICSYNC system.

A sprint planning meeting was conducted to review and prioritize the product backlog items. During this meeting, the researchers identified the most critical features to be implemented first, focusing on the Student Panel as the initial sprint goal. The team discussed the relative importance of each feature and estimated the effort required for each task. The selected features were then broken down into manageable tasks, forming the Sprint Backlog, which served as the foundation for organizing, tracking, and monitoring the work to be carried out during the sprint.

Sprint 1, after gathering the necessary information about the Admission and Enrollment process, the researchers focused on designing and developing the proposed system. The system interface was created using Figma, which served as the guide for the layout, structure, and navigation of the platform. The Admission and Enrollment module was organized into clear steps, including branch selection (Bacoor, GMA, or Taytay), entry of personal information and program or strand selection, uploading of required documents, application confirmation with tracking number, and viewing of entrance examination details for college applicants. Separate landing pages and login interfaces were also developed for different user roles such as the Student Portal, Admin Portal, and Area Manager Panel. The Student Portal allows users to register and log in using their selected branch and student number, while the Area Manager Panel displays the total number of students per branch and provides filtering options to view detailed student records. Most pages are already responsive, with minor interface elements still needing improvement. Admin login credentials are temporarily declared in the code, and the “Forgot Password” feature is not yet functional. Overall, Sprint 1 established the system’s design, structure, and user flow in preparation for the next phase of development.

Sprint 2 focused on the continued development of the system by implementing its core functionalities and improving overall system operations. After establishing the system design in Sprint 1, the researchers developed the main modules for the Area Manager, Branch Administrator, and Student using both front-end and back-end technologies.

The Area Manager dashboard was designed to provide a centralized view of data across all selected branches, including the total number of enrolled students, senior high school and college counts, and pending reports. It also includes graphical representations such as bar and pie charts to present enrollment data per course or strand. Additional features such as report management, student record viewing, staff account management, and profile settings were also implemented.

For the Branch Administrator, a dashboard was created to display student statistics within the branch, including those with complete and incomplete requirements. The system includes modules for managing student records, uploading grades using a template, and handling enrollees through new admissions, enrollment requests, and academic management. These features support application review, class assignment, subject and instructor management, and scheduling. Additional functions such as alumni records, report submission, and a trash feature for data recovery were also developed.

The Admission Portal was also implemented to allow applicants to apply online. It includes options to start a new application or continue using a tracking number. The process involves selecting student status and branch, entering personal information, uploading requirements, and receiving a tracking number to monitor application status. Approved applicants can then view entrance examination details and download their permit.

On the student side, the system allows access to profile details, grades, schedules, enrollment requests, and announcements. Students can also monitor their credential status and download necessary documents. Separate login interfaces were implemented for students and staff, requiring students to select their branch before logging in.

Overall, Sprint 2 focused on developing the system's main functionalities, strengthening role-based access, and improving data management in preparation for system testing and further refinement.

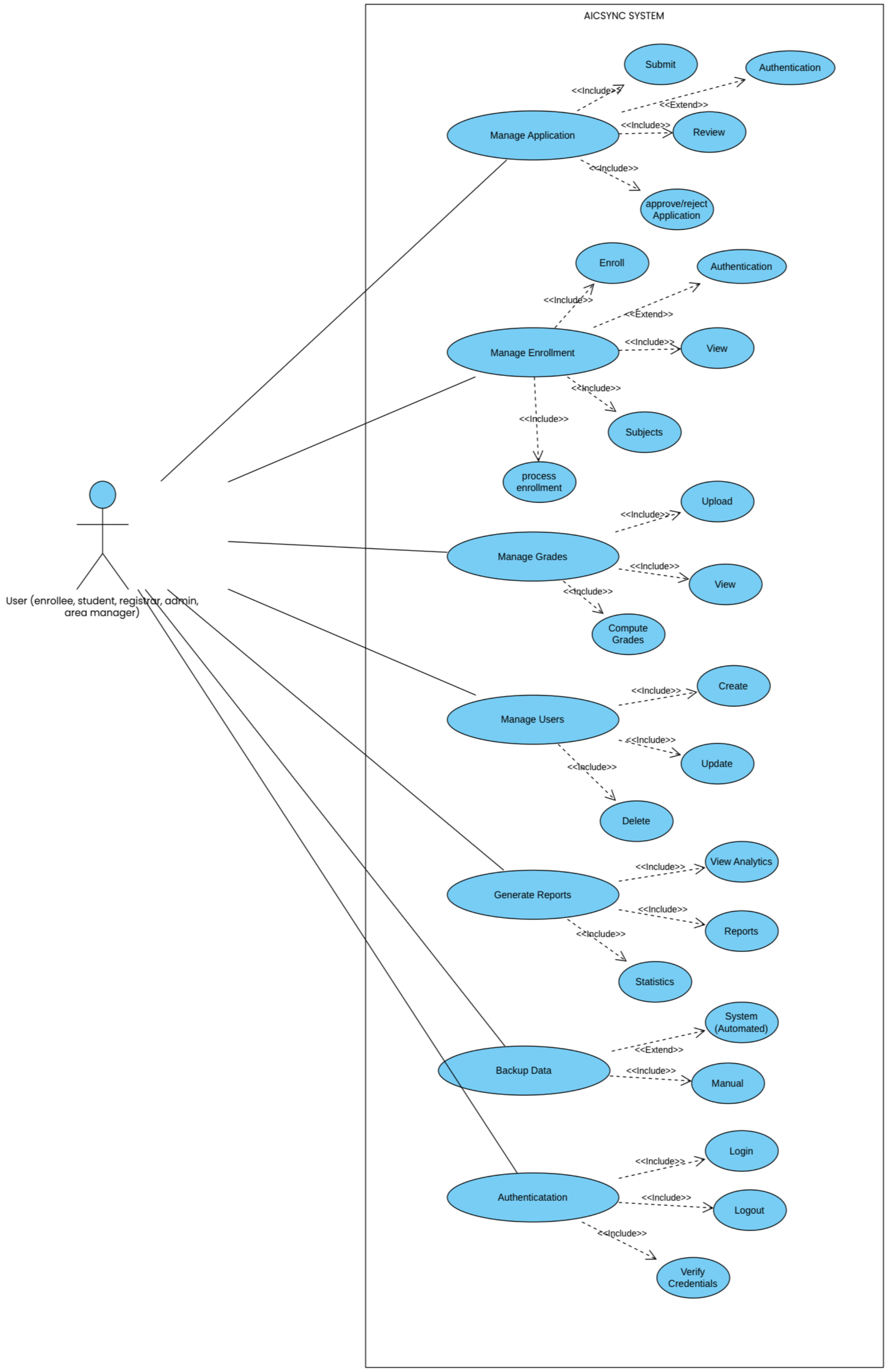

Figure 3. Shows the Use Case Diagram of AICSYNC

Centralized Multi-Branch Enrollment & Academic Records System of Asian Institute of Computer Studies. The diagram illustrates the interactions between the different system users and the various functionalities available within the system. It shows how each user role engages with the system to perform specific tasks related to enrollment, academic record management, and system administration.

The diagram identifies a general actor labeled as “User,” which represents multiple roles within the system, including the enrollee, student, registrar, administrator, and area manager. Although grouped under a single actor, each user is granted specific permissions through role-based access control, ensuring that system functionalities are accessed according to user responsibilities.

The diagram highlights several major system functions, including Manage Application, Manage Enrollment, Manage Grades, Manage Users, Generate Reports, Backup Data, and Authentication. Each of these functions represents key processes within the system that support centralized academic management.

The **Manage Application** use case covers the submission and processing of student applications. This includes actions such as submitting application forms, reviewing submitted information, and approving or rejecting applications. This process ensures that all applicant data is properly verified before proceeding to enrollment.

The **Manage Enrollment** use case focuses on handling the enrollment process. It includes enrolling students, processing enrollment details, viewing enrollment records, and managing subject assignments. This function ensures that student enrollment across multiple branches is properly recorded and organized within the system.

The **Manage Grades** use case allows authorized users to upload, view, and compute student grades. This ensures accurate recording of academic performance and provides students with access to their academic records through the system.

The **Manage Users** function includes creating, updating, and deleting user accounts. This ensures proper system access control and allows administrators to manage user roles and permissions efficiently.

The **Generate Reports** use case provides users, particularly administrators and area managers, with access to system-generated reports and analytics. These reports support monitoring of enrollment data, academic performance, and overall system activity, which are essential for decision-making.

The **Backup Data** use case ensures data security and system reliability by allowing both automatic and manual backup of system data. This feature helps prevent data loss and supports system recovery when needed.

Lastly, the **Authentication** use case handles user login, logout, and verification of credentials. This ensures that only authorized users can access the system and perform their respective tasks securely.

Overall, the Use Case Diagram provides a clear and structured representation of how users interact with the AICSYNC system. It demonstrates how the system supports different user roles while maintaining centralized control over enrollment and academic records across selected AICS branches.

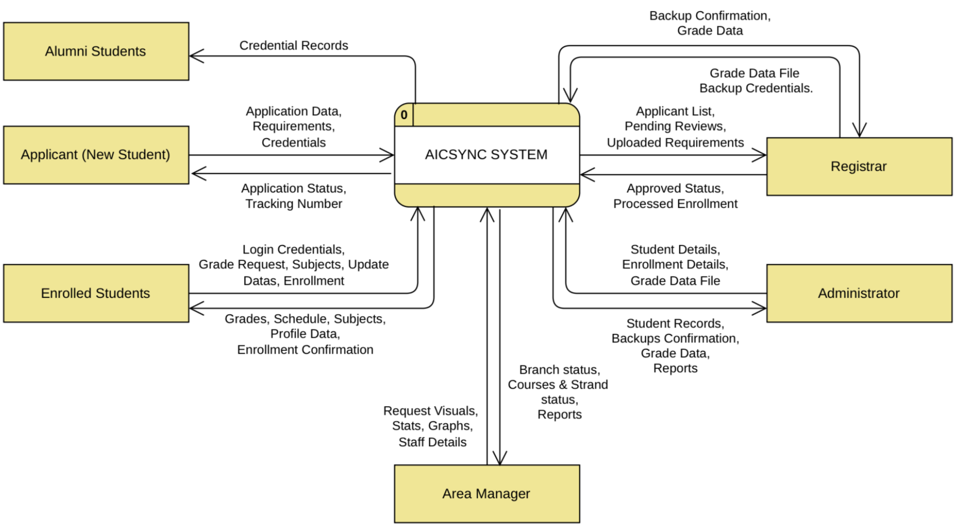

Figure 4. Level 0 of Data Flow Diagram

Figure 4 presents the Level 0 Data Flow Diagram of the AICSYNC: Centralized Multi-Branch Enrollment and Academic Records System of the Asian Institute of Computer Studies (AICS). The diagram provides a high-level overview of how the system interacts with external entities, including the Applicant/Student, Registrar, Branch Administrator, and Area Manager.

At this stage, the system is represented as a single process that receives inputs such as application data, enrollment requests, and academic records from users. In return, the system provides outputs including enrollment status, academic information, reports, and system-generated data. The diagram highlights how data flows between users and the system, emphasizing the role of AICSYNC as a centralized platform for managing enrollment and academic records across multiple branches.

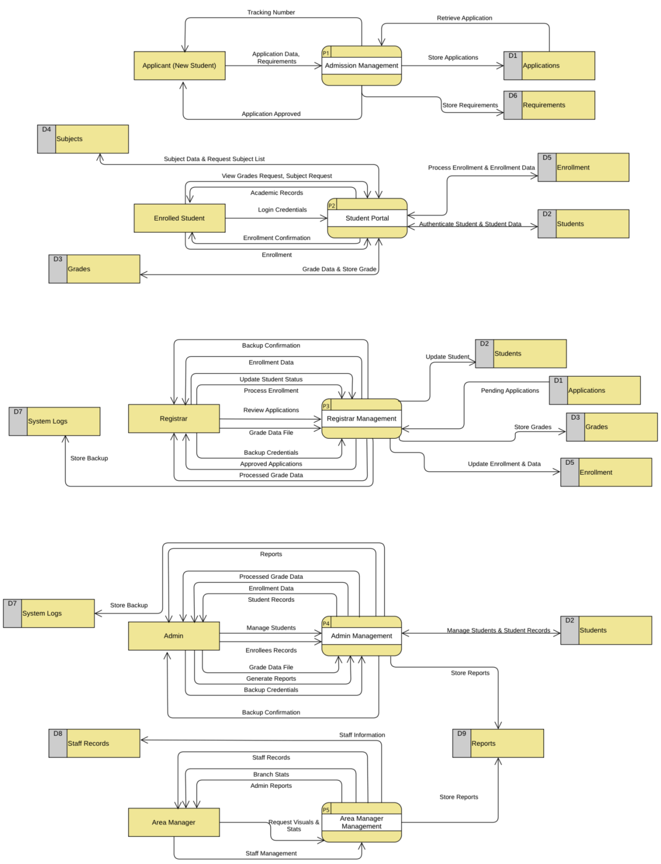

Figure 5. Level 1 of Data Flow Diagram

Figure 5 illustrates the Level 1 Data Flow Diagram for the Admission Management process of the AICSYNC system. This diagram provides a detailed view of how applicant data is collected, processed, and stored during the admission phase.

The process begins when applicants submit their personal information, selected branch, and requirements through the admission portal. The system stores this data in relevant data stores such as application records and uploaded requirements. A tracking number is generated, allowing applicants to monitor their application status. The registrar or branch administrator reviews the submitted data, and once approved, the application is forwarded for enrollment processing. This process ensures an organized and efficient handling of student applications.

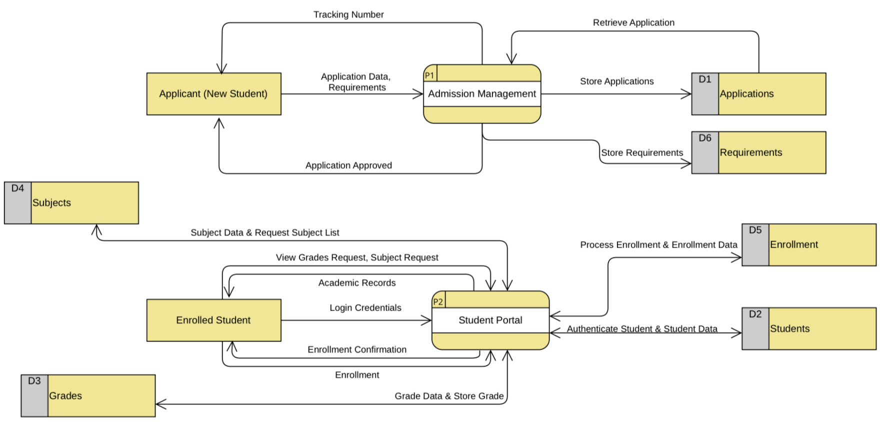

Figure 6. Level 1 of Data Flow Diagram

Figure 6 presents the Level 1 Data Flow Diagram for the Student Portal of the AICSYNC system. This diagram shows how students interact with the system after successful enrollment.

Students access the system by logging in and can perform various actions such as viewing grades, checking schedules, submitting enrollment requests, and monitoring credential status. The system retrieves and updates data from multiple data stores, including student records, grades, subjects, and enrollment data. This process ensures that students have real-time access to their academic information and can manage their academic activities efficiently through the platform.

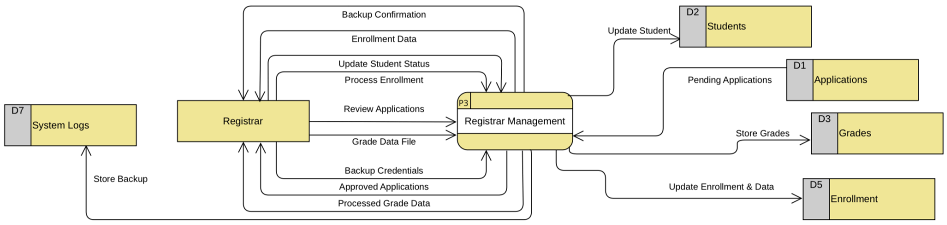

Figure 7. Level 1 of Data Flow Diagram

Figure 7 illustrates the Level 1 Data Flow Diagram for the Registrar Management process. This diagram focuses on the responsibilities of the registrar in handling academic and enrollment-related data.

The registrar reviews and processes student applications, validates submitted requirements, and updates enrollment status. The registrar is also responsible for uploading and managing student grades, ensuring that academic records are accurate and up-to-date. All processed data is stored in appropriate data stores such as student records, grades, and enrollment databases. This process plays a critical role in maintaining data accuracy and ensuring proper academic record management within the system.

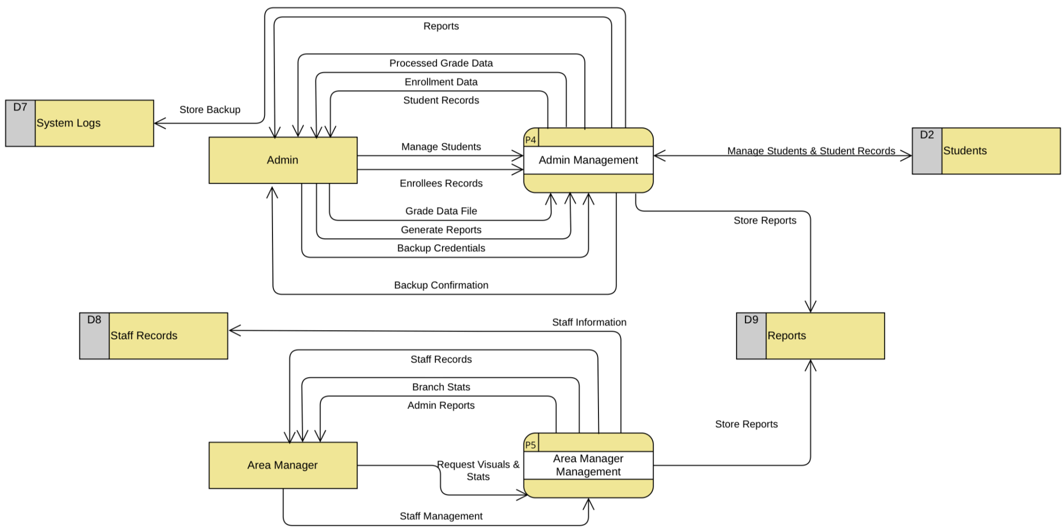

Figure 8. Level 1 of Data Flow Diagram

Figure 8 presents the Level 1 Data Flow Diagram for both the Branch Administrator and Area Manager processes. This diagram highlights how administrative users manage and monitor the system.

The Branch Administrator handles student records, enrollment data, reports, and academic management tasks within a specific branch. Meanwhile, the Area Manager oversees multiple branches by accessing consolidated data, viewing reports, and managing staff accounts. The system retrieves and stores data from various data stores, including student records, reports, and staff information. This process supports centralized monitoring, reporting, and decision-making across all covered branches of the institution.

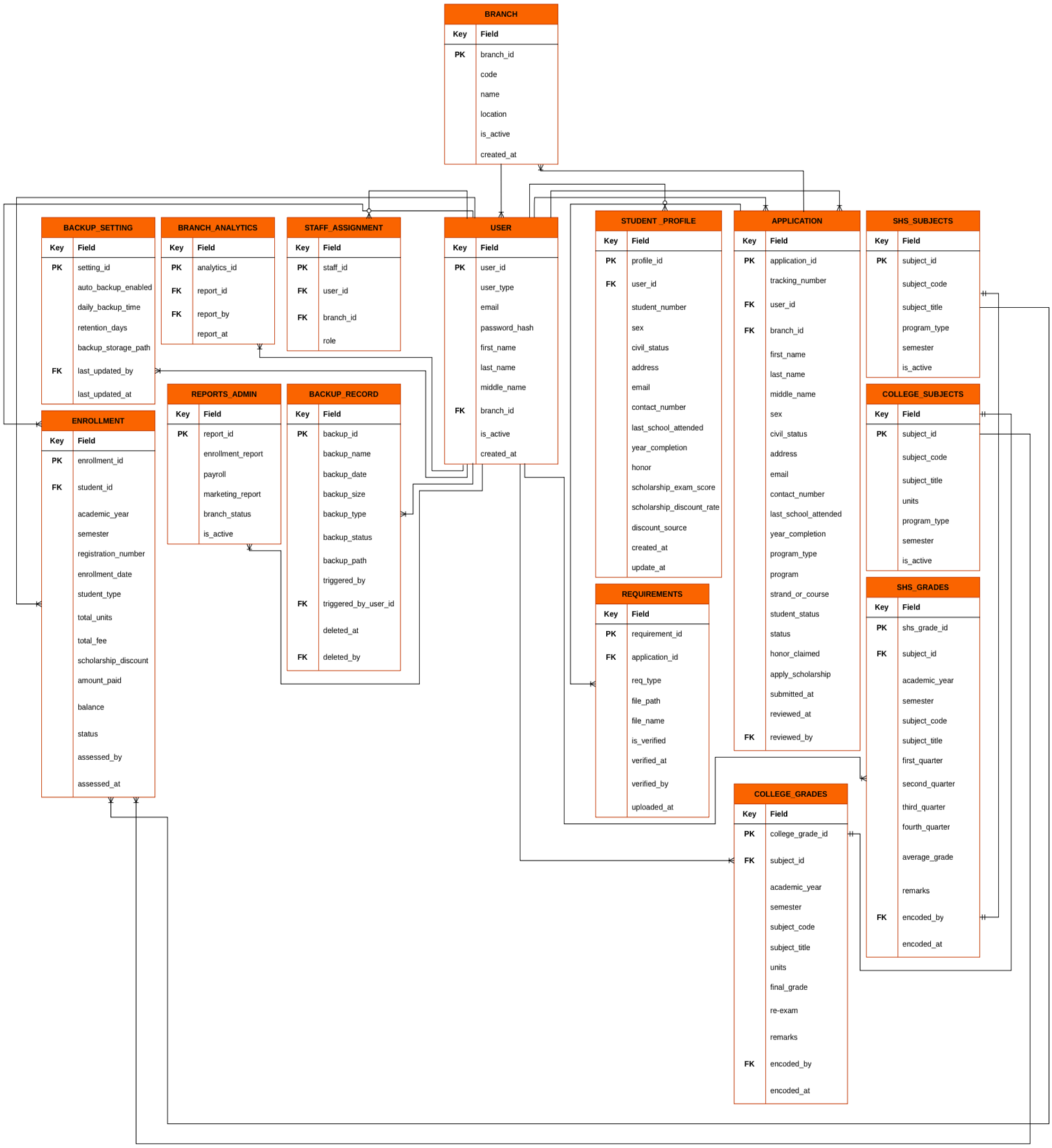

Figure 9. Entity Relationship Diagram

Figure 9 Shows the Entity Relationship Diagram (ERD) illustrates the overall database structure of the proposed system by presenting the different entities, their attributes, and the relationships among them. The central entity in the system is the User, which stores essential account information and is linked to multiple functional modules such as student profiles, applications, staff assignments, and system activities. The Branch entity defines the different campuses and is associated with users, applications, and staff assignments to support multi-branch operations. The Student Profile entity contains detailed personal and academic information of students and is connected to the Application entity, which manages enrollment data and tracks student application status. The Requirements entity is linked to applications to store and verify submitted documents. Academic components are represented through SHS Subjects, College Subjects, and their corresponding Grades entities, which manage subject details and student performance records. The Enrollment entity records student registration details, including academic year, fees, and status. Additional administrative entities such as Reports, Backup Settings, Backup Records, and Branch Analytics support system monitoring, reporting, and data management. Relationships among these entities are established through primary and foreign keys, ensuring data integrity, consistency, and efficient retrieval across the system. Overall, the ERD reflects a centralized database design that supports multi-branch enrollment, academic record management, and administrative operations.

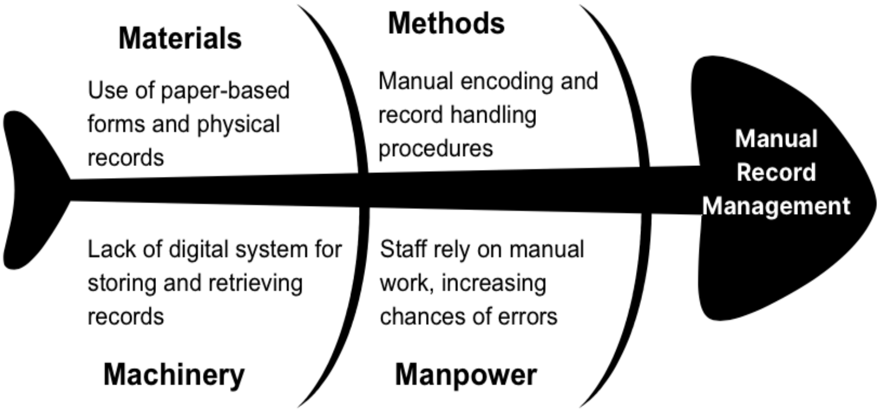

Figure 3. Fishbone diagram for problem 1

Figure 3 presents the fishbone diagram illustrating the root causes of manual record management in the existing system of the Asian Institute of Computer Studies (AICS). The diagram categorizes the causes into four major factors: materials, methods, machinery, and manpower.

Under materials, the use of paper-based forms and physical records contributes to inefficient storage and retrieval of student information. In terms of methods, manual encoding and traditional record-handling procedures increase processing time and reduce overall efficiency. The machinery factor highlights the lack of a digital system to support automated storage and retrieval, which limits accessibility and organization of records. Lastly, under manpower, staff reliance on manual work increases the likelihood of human error and inconsistencies in handling student data.

These factors collectively result in slow processing, difficulty in retrieving records, and increased risk of data loss or inconsistency. The diagram emphasizes the need for a centralized and automated system to improve record management and reduce dependency on manual processes.

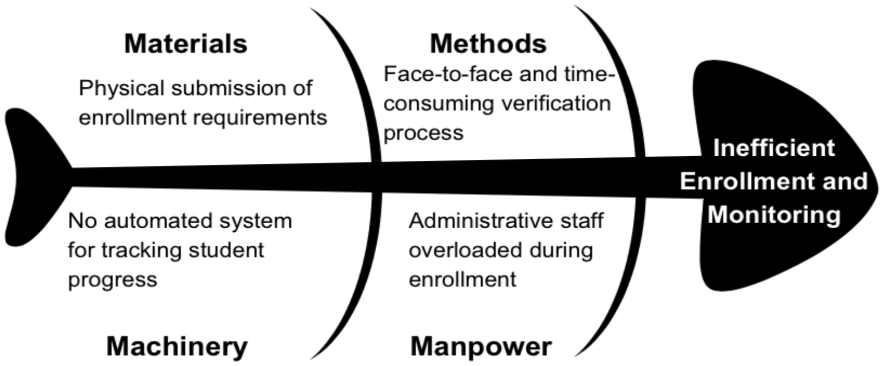

Figure 4. Fishbone diagram of the system for problem 2

Figure 4 illustrates the fishbone diagram identifying the causes of inefficient enrollment and monitoring processes within the institution. Similar to the first diagram, the causes are grouped into materials, methods, machinery, and manpower.

From the perspective of materials, enrollment requirements are submitted physically, making the process less accessible and harder to manage. In terms of methods, the reliance on face-to-face transactions and manual verification leads to time-consuming procedures and delays in processing applications. The machinery factor points to the absence of an automated system for tracking student progress, making it difficult to monitor application status and enrollment flow. Under manpower, administrative staff experience workload overload, especially during peak enrollment periods, which further slows down operations.

These combined factors result in delayed enrollment processing, inefficient monitoring of applicants, and overcrowding during transactions. The diagram highlights the need for an online and automated enrollment system to streamline processes and improve efficiency.

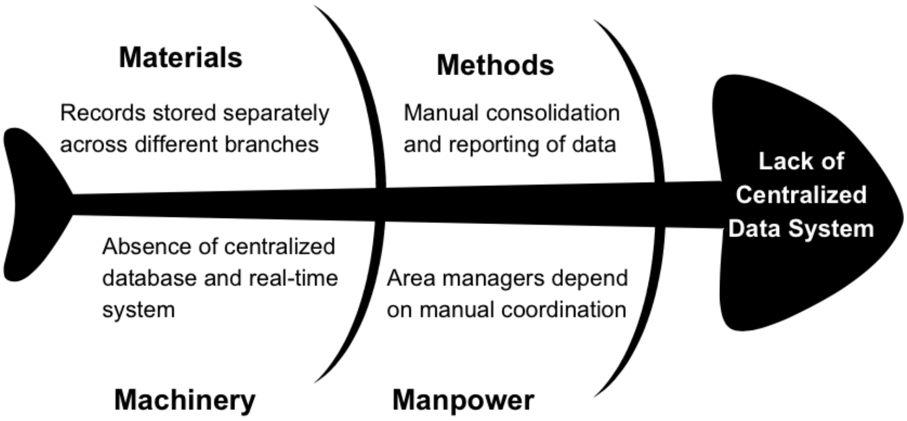

Figure 5. Fishbone diagram of the system for problem 3

Figure 5 presents the fishbone diagram showing the root causes of the lack of a centralized data system across AICS branches. The causes are again categorized into materials, methods, machinery, and manpower.

Under materials, student records are stored separately across different branches, leading to fragmented data. In terms of methods, data consolidation and report generation are performed manually, making the process slow and prone to inconsistencies. The machinery factor highlights the absence of a centralized database and real-time system, which prevents seamless data sharing across branches. Lastly, under manpower, area managers rely on manual coordination with each branch, making monitoring and decision-making inefficient.

These issues result in delayed reporting, difficulty in tracking institutional data, and lack of real-time visibility across branches. The diagram supports the need for a centralized system that integrates data from all branches to improve coordination, accuracy, and decision-making.